

# Notes: Lundborg, Plug & Rasmussen (AER, 2017)

## Can Women Have Children and a Career? IV Evidence from IVF Treatments

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# 1 Big picture

- **Question.** What is the causal effect of having a first child on women’s labor market careers?
- **Empirical challenge.** Women who have children differ from women who do not. Comparing mothers and non-mothers confounds the causal effect of childbearing with selection on preferences, health, ability, family resources, and career plans.
- **Identification idea.** The paper uses success at the first IVF treatment with embryo implantation as an instrument for childbearing among childless Danish women undergoing IVF.
- **Setting.** Denmark has rich administrative registers, universal access to many public benefits, and generous family policies. This makes the finding especially informative: even in a family-friendly institutional setting, first childbirth has large career effects.
- **Main extensive-margin result.** Having children reduces annual earnings by about DKK 70,000 in the short run, DKK 29,000 in the medium run, and DKK 31,000 in the long run. These correspond to roughly 31, 12, and 11 percent impacts.
- **Mechanism over time.** The short-run penalty is mainly labor supply: women work fewer hours and are less likely to have positive earnings around childbirth. The medium- and long-run penalty is mainly wages: women return to work but earn lower hourly wages.
- **Job reallocation channel.** Women with children appear to sort into lower-paid jobs and workplaces, and especially into jobs closer to home. The long-run commuting-distance effect is around  $-5.5$  kilometers, about a 44 percent reduction from the baseline.
- **Heterogeneity.** Earnings losses are larger for women with high own pretreatment earnings and for women with high-earning partners. Top-of-distribution women face the largest career costs.
- **Extensive versus intensive margin.** The penalty from having a first child is much stronger and more persistent than the penalty from having an additional child among women who already have children.
- **Economic takeaway.** The child penalty is not only temporary time away from work. It also changes job choice, wage trajectories, and location constraints in a persistent way.

## 2 Data and measurement

### 2.1 Danish IVF and administrative registers

- IVF register data cover fertility treatments at public and private clinics in Denmark.
- The authors merge IVF treatment records with administrative registers on demographics, education, marital status, number of children, earnings, hours, wages, occupation, workplace, sector, commuting distance, depression medication, and divorce.
- The IVF register includes 31,666 women and 96,807 IVF treatments.
- The baseline IVF sample is childless women entering their first IVF treatment with embryo implants from 1995 to 2005.
- The final baseline sample has 18,538 treated women: 13,168 with an unsuccessful first treatment and 5,370 with a successful first treatment.

**Interpretation:** The central design compares women who all reached the embryo-transfer stage of IVF. This makes the treated and comparison groups medically and behaviorally closer than a broad comparison between women who use IVF and women who do not.

## 2.2 Treatment, instrument, and outcomes

- The **instrument** is an indicator equal to one if the first IVF treatment with embryo implants leads to childbirth.
- The **endogenous treatment** is having children in year  $t$ .
- The main outcome is annual labor earnings in 2008 Danish kroner.
- Other outcomes include positive earnings, weekly hours worked, hourly wages, partner earnings, depression medication, divorce, job changes, occupation/workplace earnings, public-sector employment, and distance between home and work.
- Outcomes are summarized over three horizons: years 0–1, 2–5, and 6–10 after potential childbirth.

**Interpretation:** The horizons are economically important. The year 0–1 outcome captures maternity-leave and immediate labor-supply effects. Years 2–5 and 6–10 test whether mothers return to their pre-child career paths or face persistent wage and job-choice effects.

## 3 Models and empirical specifications

### 3.1 Baseline IV setup

The paper estimates a two-stage least squares model separately by event-time horizon:

$$F_{it} = \alpha_t X_i + \beta_t Z_i + u_{it}, \quad (1)$$

$$Y_{it} = \gamma_t X_i + \delta_t F_{it} + \nu_{it}. \quad (2)$$

- $Y_{it}$  is the labor market outcome for woman  $i$  at horizon  $t$ .
- $F_{it}$  is fertility, measured as whether the woman has children.
- $Z_i$  is IVF success at first embryo-transfer treatment.
- $X_i$  includes age at first treatment, treatment year, pretreatment education, and pretreatment outcome controls.
- $\delta_t$  is the causal effect of childbearing at horizon  $t$  for women whose fertility is shifted by first IVF success.

**Interpretation:** The design estimates a local average treatment effect for women undergoing IVF whose probability of becoming mothers is shifted by first-treatment success. It is not a generic effect for all women, but it is a clean estimate for a policy-relevant group facing infertility treatment.

### 3.2 Instrument validity conditions

The IVF instrument must satisfy:

1. **Relevance:** first IVF success must predict childbearing.
2. **Independence:** first IVF success must be unrelated to potential labor market outcomes, conditional on controls.
3. **Exclusion:** IVF success must affect labor outcomes only through childbearing, not through health, depression, divorce, or other direct channels.

**Interpretation:** The paper provides especially strong evidence for relevance. The first-stage coefficients are large and the F-statistics are far above conventional weak-instrument thresholds. The main threat is exclusion: failed IVF could affect health, mental health, or family stability. The authors address this with depression, divorce, health-proxy, and sample-selection checks.

### 3.3 Event-study reduced form

The event studies plot treatment-success coefficients for years  $-6$  to  $+10$  relative to potential childbirth, normalized to zero in period  $-1$ .

**Interpretation:** Pre-trends are a visual falsification test. If successful and unsuccessful IVF women have parallel pretreatment earnings, hours, wages, and participation, the identifying assumption becomes more credible.

## 4 Identification and empirical design

### 4.1 Why first IVF success is useful

- The first IVF treatment is used because later treatment decisions are endogenous: after failure, women choose whether to continue.
- Restricting to embryo-transfer treatments avoids comparing women whose treatments fail at earlier biological stages with those who reach the implantation stage.
- The clinical success of implanted embryos is plausibly unrelated to prior labor market outcomes after conditioning on age, treatment year, and education.

**Interpretation:** The design is strongest at the extensive margin: it asks whether women who would have remained childless after failed IVF experience different careers when first IVF success induces a first child.

### 4.2 Main identification checks

- Table 1 shows successful and unsuccessful IVF women have nearly identical pretreatment earnings and education.
- Figure 2 shows no meaningful pre-trends in earnings and most other outcomes.
- Table 2 shows near-zero pretreatment effects on baseline outcomes.
- Depression and divorce outcomes are small, reducing concern that failed IVF directly changes labor supply through mental health or marital instability.
- Sensitivity checks add medical controls and restrict to women with partner-side infertility.

**Interpretation:** The design is not a randomized trial, but the evidence supports a quasi-random interpretation of first-treatment success among women who reach embryo transfer.

### 4.3 Limits to external validity

- IVF women are older, better educated, and have higher earnings than representative mothers.
- Denmark has generous leave, job protection, and child-care policies.
- The results therefore may not directly transfer to younger mothers, countries with weaker family policy, or women not seeking fertility treatment.

**Interpretation:** These limits do not weaken the internal validity of the core IV estimate. They affect how far the estimates can be generalized.

## 5 Tables and figures

### 5.1 Figure 1 and Table 1: IVF technology and baseline comparability

**Read:**

- Figure 1 shows IVF use rising until about 2000 and stabilizing afterward.
- IVF success rates rise over 1995–2005, while the number of eggs collected and embryos transferred declines.
- This pattern points to improving IVF technology.
- Table 1 shows that successful and unsuccessful first-treatment IVF women are highly similar before treatment: pretreatment annual earnings are DKK 245.4k for failures and DKK 243.9k for successes.
- Successful women are about one year younger, which is expected because maternal age predicts IVF success.
- Compared with representative Danish mothers, IVF women are older, more educated, and have higher pretreatment earnings.

**Economic message:** The figure explains why treatment timing and technology controls matter. The table supports the key quasi-random claim: conditional on age and treatment timing, success and failure groups look similar before treatment, while IVF women as a whole are positively selected relative to representative mothers.

**Interpretation:** The most important comparison is not IVF women versus all women; it is successful versus unsuccessful first-treatment IVF women. Their pretreatment earnings are nearly the same, but posttreatment earnings diverge sharply, which motivates the causal design.

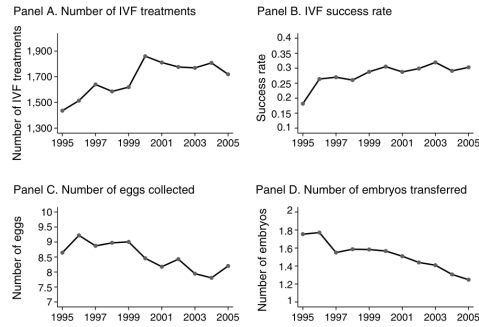


FIGURE 1. IVF TREATMENTS AND TECHNOLOGY, 1995–2005

Notes: The figure plots mean values for the period 1995–2005. The means are calculated for the sample of IVF women who entered their first treatment during 1995–2005 with no children.

(a) Figure 1: IVF treatments and technology.

|                                       | IVF failure<br>(1)   | IVF success<br>(2)   | Representative sample<br>(3) | (2)–(1)            | (2)–(3)           |
|---------------------------------------|----------------------|----------------------|------------------------------|--------------------|-------------------|
| <i>Pretreatment outcomes</i>          |                      |                      |                              |                    |                   |
| Age at first treatment                | 32.490<br>(4.445)    | 31.415<br>(3.886)    | 28.274<br>(4.297)            | –1.075<br>(0.069)  | 3.141<br>(0.060)  |
| Year at first treatment               | 2,000.149<br>(3.121) | 2,000.295<br>(3.069) | 2,001.446<br>(4.069)         | 0.146<br>(0.050)   | –1.151<br>(0.056) |
| Annual earnings (1,000s DKK)          | 245.360<br>(143.366) | 243.912<br>(131.741) | 201.717<br>(136.384)         | –1.448<br>(2.268)  | 42.195<br>(1.906) |
| Schooling                             | 12.820<br>(2.359)    | 12.843<br>(2.294)    | 12.548<br>(2.325)            | 0.023<br>(0.038)   | 0.295<br>(0.033)  |
| Partner earnings (1,000s DKK)         | 327.006<br>(209.665) | 322.318<br>(191.939) | 287.883<br>(185.995)         | –4.688<br>(3.464)  | 34.436<br>(2.722) |
| Partner schooling                     | 12.678<br>(2.389)    | 12.673<br>(2.323)    | 12.547<br>(2.316)            | –0.005<br>(0.040)  | 0.125<br>(0.034)  |
| Sickness benefits                     | 0.170<br>(0.376)     | 0.169<br>(0.375)     | 0.143<br>(0.350)             | –0.001<br>(0.006)  | 0.026<br>(0.005)  |
| Married                               | 0.521<br>(0.500)     | 0.523<br>(0.500)     | 0.306<br>(0.461)             | 0.002<br>(0.008)   | 0.217<br>(0.006)  |
| Positive earnings                     | 0.910<br>(0.288)     | 0.922<br>(0.268)     | 0.900<br>(0.300)             | 0.013<br>(0.005)   | 0.022<br>(0.004)  |
| Full-time employment <sup>a</sup>     | 0.934<br>(0.248)     | 0.934<br>(0.249)     | 0.780<br>(0.414)             | 0.000<br>(0.004)   | 0.154<br>(0.006)  |
| log (hourly wages (DKK)) <sup>a</sup> | 5.202<br>(0.340)     | 5.194<br>(0.319)     | 5.118<br>(0.430)             | 0.008<br>(0.006)   | 0.076<br>(0.006)  |
| Weekly hours worked <sup>a</sup>      | 28.620<br>(7.734)    | 28.670<br>(7.729)    | 26.626<br>(9.083)            | 0.049<br>(0.137)   | 2.044<br>(0.134)  |
| Public sector <sup>b</sup>            | 0.588<br>(0.492)     | 0.582<br>(0.493)     | 0.444<br>(0.497)             | –0.006<br>(0.008)  | 0.137<br>(0.007)  |
| <i>Posttreatment outcomes</i>         |                      |                      |                              |                    |                   |
| Annual earnings (1,000s DKK)          | 241.815<br>(144.983) | 211.525<br>(128.649) | 178.907<br>(127.712)         | –30.290<br>(2.274) | 32.618<br>(1.788) |
| Positive earnings                     | 0.888<br>(0.267)     | 0.864<br>(0.282)     | 0.852<br>(0.349)             | –0.024<br>(0.004)  | 0.011<br>(0.005)  |
| Log (hourly wages (DKK)) <sup>a</sup> | 5.314<br>(0.317)     | 5.294<br>(0.316)     | 5.255<br>(0.323)             | –0.020<br>(0.006)  | 0.038<br>(0.005)  |
| Hours worked <sup>a</sup>             | 30.034<br>(26.823)   | 27.216<br>(21.634)   | 25.390<br>(7.655)            | –2.818<br>(0.483)  | 1.826<br>(0.151)  |
| Observations                          | 13,168               | 5,370                | 103,826                      |                    |                   |

Notes: The table shows descriptive statistics for three samples: (i) women having an unsuccessful first IVF treatment; (ii) women having a successful first IVF treatment; (iii) and a representative sample of Danish women who had their first child born during the study period. Columns 1 to 3 show means with standard deviations in parentheses. Column 4 shows the difference in means between columns 2 and 1 and column 5 shows the corresponding difference between columns 2 and 3. Annual earnings are reported in 2008 Danish Kroner (DKK 100 corresponds to US\$20 as of August 2008).

<sup>a</sup> Means conditional on having positive earnings.

<sup>b</sup> Means conditional on observing sector of work.

between successfully and unsuccessfully treated women, and between first time successfully treated IVF mothers and representative mothers.

We can make two informative comparisons. First, we compare successfully

(b) Table 1: Descriptive statistics.

## 5.2 Figure 2 and Table 2: first stage and reduced form

Read:

- Figure 2 Panel A shows a sharp first-stage effect: first IVF success raises the probability of having children immediately and the effect remains positive for ten years.
- Panel B shows a large drop in annual earnings after successful treatment, with the largest fall around childbirth and persistent negative effects later.
- Panels C and D show short-run declines in participation and hours, but those effects fade.
- Panel E shows that log hourly wages fall in the medium and long run.
- Panels G and H show little evidence that depression or divorce are the main drivers.
- Table 2 quantifies the reduced form: IVF success raises fertility by 0.694 in years 0–1, 0.320 in years 2–5, and 0.227 in years 6–10.
- The earnings reduced form is –48,633, –9,402, and –6,960 DKK across the three horizons.

**Economic message:** The reduced form shows the raw treatment-success effect before scaling by the first stage. The fading first stage is important: unsuccessful first-treatment women may eventually have children through later treatments, which reduces the long-run first-stage gap.

**Interpretation:** The temporal pattern is the paper’s key mechanism. The short-run earnings penalty follows hours and participation, but the long-run penalty follows wages and job sorting. This implies that the first child changes the quality or type of work, not only the quantity of work.

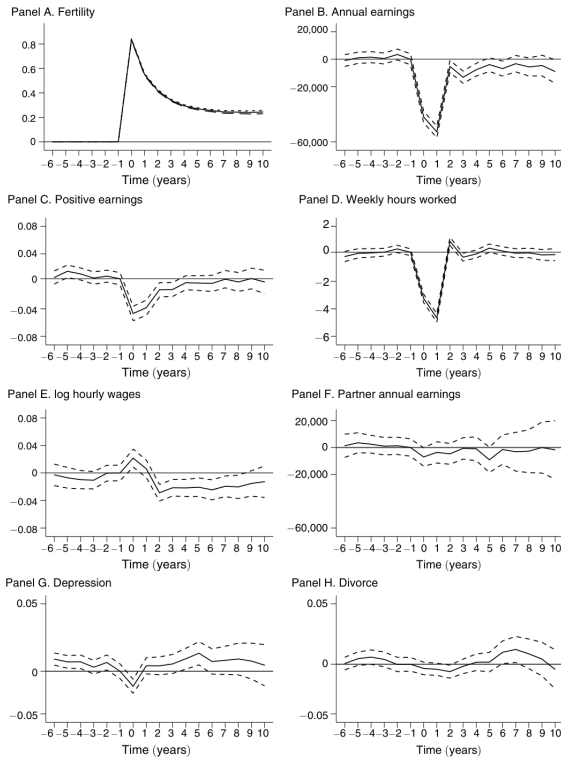


FIGURE 2. EVENT STUDY GRAPHS OF IVF TREATMENT EFFECTS ON VARIOUS OUTCOMES

Notes: The figures plot coefficients from an event-study analysis. Event time is defined as years before and after (potential) childbirth. The coefficients in period  $-1$  are normalized to 0. The models are estimated for the sample of IVF treated women who had their first IVF treatment between 1995 and 2005. The coefficients in panel F are estimated for the partners of the IVF-treated women. For details on the model, see the text.

TABLE 2—IVF TREATMENT EFFECTS ON FEMALE LABOR OUTCOMES: RESULTS FROM FIRST-STAGE AND REDUCED-FORM REGRESSIONS

| Independent variable       | Fertility (1)    | Earnings (2)       | Positive earnings (3) | Weekly hours (4)  | Wages (5)         | log wages (6)     | Partner earnings (7) | Depression (8)    | Divorce (9)       |
|----------------------------|------------------|--------------------|-----------------------|-------------------|-------------------|-------------------|----------------------|-------------------|-------------------|
| <i>Panel A: Years 0–1</i>  |                  |                    |                       |                   |                   |                   |                      |                   |                   |
| IVF success                | 0.694<br>(0.004) | −48,633<br>(1,439) | −0.050<br>(0.004)     | −4,036<br>(0.131) | 2,899<br>(2,212)  | 0.009<br>(0.005)  | −5,375<br>(2,470)    | −0.013<br>(0.003) | −0.009<br>(0.002) |
| Observations               | 18,538           | 18,538             | 18,538                | 14,022            | 14,022            | 14,022            | 16,689               | 18,538            | 18,538            |
| F-statistic                | 38,427           |                    |                       |                   |                   |                   |                      |                   |                   |
| <i>Panel B: Years 2–5</i>  |                  |                    |                       |                   |                   |                   |                      |                   |                   |
| IVF success                | 0.320<br>(0.004) | −9,402<br>(1,703)  | −0.013<br>(0.004)     | 0.476<br>(0.114)  | −8,690<br>(1,437) | −0.034<br>(0.005) | −3,523<br>(3,004)    | 0.002<br>(0.003)  | −0.009<br>(0.003) |
| Observations               | 18,435           | 18,435             | 18,435                | 12,332            | 12,332            | 12,332            | 16,590               | 18,435            | 18,435            |
| F-statistic                | 6,281            |                    |                       |                   |                   |                   |                      |                   |                   |
| <i>Panel C: Years 6–10</i> |                  |                    |                       |                   |                   |                   |                      |                   |                   |
| IVF success                | 0.227<br>(0.005) | −6,960<br>(2,397)  | −0.003<br>(0.005)     | 0.103<br>(0.134)  | −5,348<br>(1,861) | −0.021<br>(0.006) | −5,082<br>(4,536)    | 0.003<br>(0.005)  | 0.003<br>(0.005)  |
| Observations               | 13,779           | 13,779             | 13,779                | 9,627             | 9,627             | 9,627             | 12,367               | 13,779            | 13,779            |
| F-statistic                | 2,273            |                    |                       |                   |                   |                   |                      |                   |                   |
| Baseline mean              | —                | 223,038            | 0.90                  | 28.63             | 183.01            | 5.16              | 301,683              | 0.05              | 0.05              |
| Pretreatment effect        | —                | 874<br>(1,811)     | 0.010<br>(0.004)      | 0.519<br>(0.375)  | −0.061<br>(1.162) | 0.001<br>(0.005)  | −1,298<br>(2,800)    | −0.005<br>(0.003) | 0.001<br>(0.003)  |

Notes: This table shows first-stage and reduced-form regression estimates on the effect of IVF treatment success (0/1) on various outcomes measured at  $t = 0, 1, 2, 5$ , and  $6-10$ . In column 1, the coefficient represents the effect of IVF success on the probability of having children during the time period considered. In the reduced-form regressions, the coefficient represents the effect of IVF success on the average of the outcome during the time period considered. Time period  $t = 0$  refers to the year of the (potential) child birth. All regressions control for age at first IVF treatment, year of first IVF treatment, pretreatment education, and the pretreatment average of the outcome studied taken over years 1–4 before the first IVF treatment. There are two exceptions. The first-stage regression does not include the pretreatment average because it is zero by construction. The depression regression includes the pretreatment value at  $t = 1$  because data on antidepressants are only complete from 1994 onward. The  $F$ -statistics in the table refers to  $F$ -tests of the significance of the instrument in the first-stage regressions. The baseline mean refers to the mean of the outcome taken over years 1–4 before the first IVF treatment (for the sample observed 0–1 years after the year of the (potential) childbirth). The pretreatment effect refers to the reduced-form effect of success at first IVF treatment on the pretreatment baseline mean. Robust standard errors are in parentheses.

rate during the first two years following a successful treatment, but negative effects in the long run. In columns 5 and 6, we find that hourly wages fall with 2 to 3 percent in the medium and long run.<sup>8</sup>

### 5.3 Table 3: IV estimates at the extensive margin

Read:

- Having children reduces annual earnings by DKK 70,088 in years 0–1, DKK 29,378 in years 2–5, and DKK 30,675 in years 6–10.
- Percent impacts are  $-31\%$ ,  $-12\%$ , and  $-11\%$ .
- Positive earnings decline by 7.2 percentage points in the short run, 4.1 points in the medium run, and 1.5 points in the long run.
- Weekly hours fall by 5.9 hours in the short run, but hours are not lower in the medium and long run.
- Wages are not lower in the short run, but fall by about DKK 27 in the medium run and DKK 25 in the long run.

**Economic message:** The first child generates a large and persistent earnings penalty. The source changes over time: initially, mothers work less; later, they work but at lower hourly wages.

**Interpretation:** The long-run wage result is the central economic finding. It rules out a story in which the child penalty is only maternity leave or temporary labor-force exit. It suggests durable changes in career path, job match, occupation, workplace, bargaining position, or flexibility constraints.

TABLE 3—FERTILITY EFFECTS ON FEMALE LABOR MARKET OUTCOMES:  
RESULTS FROM INSTRUMENTAL VARIABLE REGRESSIONS

|                            | Earnings<br>(1)     | Positive<br>earnings<br>(2) | Weekly<br>hours<br>(3) | Wages<br>(4)       |
|----------------------------|---------------------|-----------------------------|------------------------|--------------------|
| <i>Panel A. Years 0–1</i>  |                     |                             |                        |                    |
| Fertility                  | −70,088<br>(2,054)  | −0.072<br>(0.006)           | −5.911<br>(0.190)      | 4,244<br>(3,235)   |
| Percent impact             | −31                 | −8                          | −21                    | 2                  |
| Observations               | 18,538              | 18,538                      | 14,022                 | 14,022             |
| <i>Panel B. Years 2–5</i>  |                     |                             |                        |                    |
| Fertility                  | −29,378<br>(5,285)  | −0.041<br>(0.012)           | 1.473<br>(0.355)       | −26,851<br>(4,453) |
| Percent impact             | −12                 | −5                          | 5                      | −13                |
| Observations               | 18,435              | 18,435                      | 12,332                 | 12,332             |
| <i>Panel C. Years 6–10</i> |                     |                             |                        |                    |
| Fertility                  | −30,675<br>(10,546) | −0.015<br>(0.022)           | 0.487<br>(0.634)       | −25,301<br>(8,801) |
| Percent impact             | −11                 | −2                          | 2                      | −12                |
| Observations               | 13,779              | 13,779                      | 9,627                  | 9,627              |
| Baseline mean              | 223,038             | 0.90                        | 28.63                  | 183.01             |
| Pretreatment effect        | 874<br>(1,811)      | 0.010<br>(0.004)            | 0.519<br>(0.375)       | −0.061<br>(1.162)  |

*Notes:* This table shows instrumental variable estimates of the effect of having children on various labor market outcomes at  $t = 0-1$ , 2–5, and 6–10. The coefficients represent the effect of having children. Having children in year  $t$  is instrumented with IVF treatment success. Time period  $t = 0$  refers to the year of (potential) childbirth. All regressions control for age at first IVF treatment, year of first IVF treatment, pretreatment education, and the pretreatment average of the outcome studied taken over years 1–4 before the first IVF treatment. The baseline mean refers to the average of the outcome taken over years 1–4 before the first IVF treatment. The pretreatment effect refers to the reduced-form effect of success at first IVF treatment on pretreatment baseline mean. Robust standard errors are in parentheses.

In column 1, we take annual labor earnings as our labor market outcome, fertility measured at the extensive margin as our endogenous variable, success at first IVF treatment as our instrument, and IVF treatment year, the age of the mother at the first

TABLE 4—FERTILITY EFFECTS ON JOB CHANGES AND JOB CHARACTERISTICS:  
RESULTS FROM INSTRUMENTAL VARIABLE REGRESSIONS

|                            | Occ.<br>change<br>(1) | Firm<br>change<br>(2) | Occ.<br>earnings<br>(3) | Occ.<br>(log)<br>wages<br>(4) | Occ.<br>hours<br>(5) | Firm<br>earnings<br>(6) | Firm<br>(log)<br>wages<br>(7) | Firm<br>hours<br>(8) | Firm<br>gender<br>ratio<br>(9) | Public<br>sector<br>(10) | Distance<br>(in km)<br>(11) |
|----------------------------|-----------------------|-----------------------|-------------------------|-------------------------------|----------------------|-------------------------|-------------------------------|----------------------|--------------------------------|--------------------------|-----------------------------|
| <i>Panel A. Years 0–1</i>  |                       |                       |                         |                               |                      |                         |                               |                      |                                |                          |                             |
| Fertility                  | 0.017<br>(0.010)      | 0.040<br>(0.010)      | −3.687<br>(1.090)       | 0.001<br>(0.002)              | −0.346<br>(0.080)    | −1.197<br>(1.003)       | −0.002<br>(0.003)             | −0.050<br>(0.061)    | −0.008<br>(0.003)              | −0.005<br>(0.005)        | 1.379<br>(0.568)            |
| Observations               | 17,941                | 18,194                | 18,019                  | 18,019                        | 18,019               | 18,264                  | 18,195                        | 18,194               | 18,280                         | 17,914                   | 14,440                      |
| <i>Panel B. Years 2–5</i>  |                       |                       |                         |                               |                      |                         |                               |                      |                                |                          |                             |
| Fertility                  | 0.046<br>(0.028)      | 0.038<br>(0.025)      | −4.881<br>(2,679)       | −0.005<br>(0.006)             | −0.262<br>(0.179)    | −5.761<br>(2,528)       | −0.017<br>(0.008)             | 0.213<br>(0.034)     | −0.004<br>(0.008)              | 0.010<br>(0.013)         | −3.667<br>(1.083)           |
| Observations               | 14,166                | 18,058                | 17,934                  | 17,934                        | 17,934               | 18,169                  | 18,072                        | 18,084               | 18,188                         | 17,824                   | 10,618                      |
| <i>Panel C. Years 6–10</i> |                       |                       |                         |                               |                      |                         |                               |                      |                                |                          |                             |
| Fertility                  | 0.010<br>(0.044)      | −0.013<br>(0.043)     | −5.030<br>(4,962)       | −0.005<br>(0.011)             | −0.383<br>(0.315)    | −7.063<br>(4,839)       | −0.022<br>(0.014)             | −0.301<br>(0.366)    | 0.001<br>(0.016)               | 0.023<br>(0.026)         | −5.530<br>(2,145)           |
| Observations               | 11,680                | 11,782                | 13,470                  | 13,470                        | 13,470               | 13,595                  | 13,522                        | 10,974               | 13,616                         | 13,337                   | 5,173                       |
| Baseline mean              | 0.54                  | 0.57                  | 221,346                 | 5.24                          | 22.33                | 244,288                 | 5.20                          | 25.58                | 0.63                           | 0.55                     | 12.58                       |
| Pretreatment effect        | −0.005<br>(0.008)     | −0.007<br>(0.008)     | −270<br>(990)           | −0.001<br>(0.002)             | 0.125<br>(0.070)     | 955<br>(993)            | −0.004<br>(0.003)             | 0.022<br>(0.064)     | −0.002<br>(0.003)              | −0.009<br>(0.008)        | 0.639<br>(0.360)            |

*Notes:* This table shows instrumental variable estimates of the effect of having children on various labor market outcomes at  $t = 0-1$ , 2–5, and 6–10. The coefficients represent the effect of having children. Having children in year  $t$  is instrumented with IVF treatment success. In columns 1 and 2, the outcome is a binary indicator taking the value of 1 if the woman was recorded as changing occupation or workplace in the short, medium, and long run. Time period  $t = 0$  refers to the year of the (potential) childbirth. All regressions control for age at first IVF treatment, year of treatment, pretreatment education, and pretreatment outcomes in years 1–4 before the first IVF treatment. The baseline mean refers to the average of the outcome taken over years 1–4 before the first IVF treatment. The pretreatment effect refers to the reduced-form effect of success at first IVF treatment on pretreatment baseline mean. Robust standard errors are in parentheses.

significant in the short run. In columns 3 and 6 we find that women who change occupation and workplace because of children also sort into occupations and workplaces with on average lower earnings. In addition, we find that once these women start working in occupations and workplaces that pay less, they keep on working in such occupations and workplaces. The long-run estimate for the occupation earnings average suggests that women, six to ten years after giving birth, work in occupations in which the average female worker earns approximately DKK 5,000 less, which is about 2 percent of the occupation earnings average in the sample.<sup>14</sup> Although the estimates are all small and not always precise enough to be statistically significant.

## 5.4 Table 4: job changes, job characteristics, and commuting distance

Read:

- Occupation and firm changes increase modestly after childbearing, especially in the medium run.
- Mothers sort into occupations and firms with lower average earnings.
- Effects on public-sector work, female-dominated firms, and hours characteristics are mostly small.
- The standout margin is distance to work: the effect is −3.667 km in years 2–5 and −5.530 km in years 6–10.
- With a baseline distance of 12.58 km, the long-run effect is about a 44 percent reduction.

**Economic message:** The job-characteristic evidence points toward family-friendly location constraints rather than just lower effort. Mothers appear to choose jobs closer to home, and those jobs pay less.

**Interpretation:** This table is the bridge between earnings losses and mechanisms. It suggests the child penalty arises from a reoptimization of job attributes: proximity and family compatibility become more valuable, and wages fall as compensation for those nonwage attributes.

## 5.5 Table 5 and Figure 3: heterogeneity

Read:

- High-earning women lose more in levels: DKK 86,284 in years 0–1, DKK 48,647 in years 2–5, and DKK 60,621 in years 6–10.
- Women with high-earning partners experience especially large medium- and long-run losses: DKK 79,865 and DKK 99,870.



- Public-sector women experience smaller short-run losses, consistent with more generous paid maternity leave, but medium- and long-run differences are much smaller.
- Figure 3 shows all earnings-quintile effects are negative; the largest long-run losses occur toward the top of the own or partner earnings distribution.

**Economic message:** The child penalty is not restricted to low-skill or weakly attached workers. High-potential women face especially large level losses, which implies that childbirth can reduce aggregate utilization of high human capital.

**Interpretation:** The partner-earnings results are especially informative. High partner earnings may relax household budget constraints and make it easier for mothers to choose lower-paid, closer-to-home, or more flexible jobs. The pattern is therefore consistent with household specialization and bargaining models.

TABLE 5—FERTILITY EFFECTS ON FEMALE LABOR EARNINGS: HETEROGENEITY ANALYSES

| Indicator <i>I</i>                                                          | Earnings<br>Qrt. 4<br>(1) | Schooling<br>≥ 15 yrs<br>(2) | Age<br>≥ 32 yrs<br>(3) | Partner<br>earnings<br>Qrt. 4<br>(4) | Sector<br>public<br>(5) | Time<br>≥ 2002<br>(6) |
|-----------------------------------------------------------------------------|---------------------------|------------------------------|------------------------|--------------------------------------|-------------------------|-----------------------|
| <i>Panel A. Fertility effects in reference group sample (I = 0) (years)</i> |                           |                              |                        |                                      |                         |                       |
| 0–1                                                                         | –64,033<br>(2,215)        | –68,039<br>(2,300)           | –63,904<br>(3,019)     | –64,411<br>(2,343)                   | –76,627<br>(3,353)      | –66,553<br>(2,542)    |
| 2–5                                                                         | –20,418<br>(5,383)        | –26,245<br>(5,479)           | –16,392<br>(8,890)     | –15,600<br>(5,896)                   | –36,637<br>(8,726)      | –30,600<br>(6,483)    |
| 6–10                                                                        | –19,238<br>(10,520)       | –28,538<br>(10,233)          | –16,313<br>(19,598)    | –13,963<br>(11,719)                  | –35,523<br>(16,940)     | –31,324<br>(11,647)   |
| Percent impact                                                              |                           |                              |                        |                                      |                         |                       |
| 0–1                                                                         | –35                       | –35                          | –33                    | –31                                  | –33                     | –30                   |
| 2–5                                                                         | –10                       | –12                          | –8                     | –7                                   | –15                     | –13                   |
| 6–10                                                                        | –8                        | –12                          | –6                     | –5                                   | –13                     | –12                   |
| <i>Panel B. Fertility effects in indicator group sample (I = 1) (years)</i> |                           |                              |                        |                                      |                         |                       |
| 0–1                                                                         | –86,284<br>(4,707)        | –74,280<br>(4,294)           | –74,282<br>(2,772)     | –89,586<br>(4,804)                   | –67,281<br>(2,638)      | –75,023<br>(3,410)    |
| 2–5                                                                         | –48,647<br>(13,956)       | –36,130<br>(13,274)          | –36,733<br>(6,513)     | –79,865<br>(14,666)                  | –24,212<br>(6,817)      | –28,442<br>(8,909)    |
| 6–10                                                                        | –60,621<br>(29,989)       | –31,692<br>(32,502)          | –39,166<br>(11,999)    | –99,870<br>(31,571)                  | –35,383<br>(14,030)     | –29,683<br>(23,910)   |
| Percent impact                                                              |                           |                              |                        |                                      |                         |                       |
| 0–1                                                                         | –24                       | –24                          | –30                    | –31                                  | –28                     | –32                   |
| 2–5                                                                         | –13                       | –12                          | –14                    | –27                                  | –10                     | –11                   |
| 6–10                                                                        | –16                       | –9                           | –14                    | –31                                  | –13                     | –11                   |
| <i>Panel C. Differences between the two fertility effects (years)</i>       |                           |                              |                        |                                      |                         |                       |
| 0–1                                                                         | –22,251<br>(5,202)        | –6,241<br>(4,872)            | –10,377<br>(4,107)     | –25,175<br>(5,345)                   | 9,346<br>(4,268)        | –8,471<br>(4,254)     |
| 2–5                                                                         | –28,228<br>(14,930)       | –9,885<br>(5,479)            | –20,342<br>(11,022)    | –64,265<br>(15,808)                  | 12,426<br>(11,074)      | 2,159<br>(11,019)     |
| 6–10                                                                        | –41,383<br>(31,781)       | –3,153<br>(34,075)           | –22,853<br>(22,980)    | –85,907<br>(33,676)                  | 140<br>(21,996)         | 1,640<br>(26,597)     |
| Reference mean                                                              | 182,382                   | 205,626                      | 185,204                | 205,772                              | 245,156                 | 218,403               |
| Indicator mean                                                              | 364,445                   | 258,029                      | 249,279                | 280,355                              | 220,329                 | 229,016               |

Notes: Panels A and B show instrumental variable regressions on the effect of having a child on earnings at  $t = 0-1$ , 2–5, and 6–10 on subsamples split by pretreatment earnings, education, age, partner pretreatment earnings, sector, and timing of (potential) birth. Year 0 refers to the year of the (potential) childbirth. All regressions control for age at first IVF treatment, year of first IVF treatment, pretreatment education, and earnings. Panel C shows the difference between the subsample instrumental variable estimates. The means reported at the bottom part refer to outcome averages taken over years 1–4 before the first IVF treatment in reference group and indicator group samples. Robust standard errors are in parentheses.

by age at treatment: that is, women older than 32 reduce their annual labor earnings

(a) Table 5: heterogeneity by pretreatment characteristics.

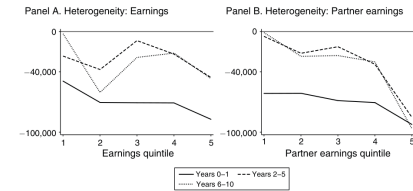


FIGURE 3. HETEROGENEITY ANALYSES

Notes: The figure plots coefficients from regressions on the effect of childbirth on annual earnings across the pretreatment distribution of earning and partner earnings. The pretreatment characteristics are measured as percentile ranks of the pretreatment variable under consideration. The coefficients represent regression estimates on sample means.

(b) Figure 3: heterogeneity across earnings quintiles.

## 5.6 Table 6: extensive versus intensive fertility margins

Read:

- Column 1 restates the extensive-margin first-child effect: –70,088, –29,378, and –30,675 DKK.
- Column 2 uses IVF success among women who already have children. The intensive-margin effect is –52,686 DKK in the short run but only –9,477 in the medium run and statistically small in the long run.

- Columns 3 and 4 use twins at first birth as the instrument. Effects are much smaller: around  $-7\%$  to  $-8\%$  in the short run and close to zero in the longer run.
- IVF and representative twin samples produce similar intensive-margin estimates, supporting some external validity for the intensive-margin comparison.

**Economic message:** The first child changes a woman’s career much more than an additional child. The extensive margin is where the persistent child penalty is concentrated.

**Interpretation:** This table sharpens the paper’s contribution relative to twin-birth studies. Twin instruments mostly identify the effect of an extra child among mothers. This paper shows that the first transition into motherhood is the larger and more durable shock to careers.

TABLE 6—FERTILITY EFFECTS ON FEMALE LABOR EARNINGS: INTENSIVE MARGINS

| Instrument:<br>Sample:     | Extensive margin                 | Intensive margin                 |                            |                              |
|----------------------------|----------------------------------|----------------------------------|----------------------------|------------------------------|
|                            | IVF success<br>IVF sample<br>(1) | IVF success<br>IVF sample<br>(2) | Twins<br>IVF sample<br>(3) | Twins<br>Repr. sample<br>(4) |
| <i>Panel A. Years 0–1</i>  |                                  |                                  |                            |                              |
| Fertility                  | −70,088<br>(2,054)               | −52,686<br>(3,198)               | −14,507<br>(2,930)         | −13,052<br>(2,197)           |
| Percent impact             | −31                              | −24                              | −7                         | −8                           |
| Observations               | 18,538                           | 4,598                            | 4,557                      | 103,826                      |
| <i>Panel B. Years 2–5</i>  |                                  |                                  |                            |                              |
| Fertility                  | −29,378<br>(5,285)               | −9,477<br>(5,151)                | −6,049<br>(5,661)          | −3,824<br>(4,574)            |
| Percent impact             | −12                              | −4                               | −3                         | −2                           |
| Observations               | 18,435                           | 4,581                            | 4,540                      | 103,178                      |
| <i>Panel C. Years 6–10</i> |                                  |                                  |                            |                              |
| Fertility                  | −30,675<br>(10,546)              | 4,518<br>(7,434)                 | −13,154<br>(9,935)         | −2,648<br>(9,938)            |
| Percent impact             | −11                              | 2                                | −5                         | −1                           |
| Observations               | 13,779                           | 3,290                            | 3,543                      | 72,987                       |
| Baseline mean              | 223,038                          | 210,064                          | 221,786                    | 174,156                      |
| Pretreatment effect        | 874<br>(1,811)                   | 6,794<br>(3,816)                 | −4,616<br>(3,592)          | 12,073<br>(2,767)            |

*Notes:* Column 1 replicates the main IV findings of Table 3. Column 2 shows the effect for women with children entering their first IVF treatment. Column 3 shows the effect of family size on earnings for women who had a successful first IVF treatment. Average family size at  $t = 0-1$ , 2–5, and 6–10 is instrumented with an indicator of having a singleton versus twin birth at first birth. Column 4 shows the corresponding results for a sample of representative, non-IVF-treated women. Time period  $t = 0$  refers to the year of the (potential) childbirth. All regressions control for age at first IVF treatment, year of first IVF treatment, pretreatment education, and earnings. The baseline mean refers to the average of the outcome variable taken over years 1–4 before the year of the first IVF treatment. The pretreatment effect refers to the reduced-form effect of success at first IVF treatment (columns 1 and 2), or of having a twin birth (columns 3 and 4), on the pretreatment baseline mean. Robust standard errors are in parentheses.

## 6 Short summary

- The paper estimates the causal effect of having a first child on women’s careers using first IVF success as an instrument.
- The design focuses on childless Danish women who reach the embryo-transfer stage of their first IVF treatment.
- Successful and unsuccessful first-treatment IVF women look very similar before treatment, especially in earnings and education.
- The first stage is strong: IVF success sharply raises the probability of having children.
- Having a first child lowers annual earnings by about 31 percent in the short run and about 11–12 percent in the medium and long run.
- Short-run losses are mainly due to labor supply and labor-force participation.
- Medium- and long-run losses are mainly due to hourly wage declines and job sorting.
- Mothers sort into jobs closer to home, which likely trades wages for family-compatible job attributes.
- Earnings losses are larger for women with high own or partner earnings.
- Additional children have smaller and less persistent effects than the first child.

- The evidence implies that even family-friendly Denmark has a large and persistent motherhood career penalty.

## **7 Expanded conclusion**

### **7.1 Core conclusion**

The paper provides strong causal evidence that having a first child imposes a large, persistent earnings penalty on women. The effect is not merely a temporary maternity-leave shock. It persists long after women return to work, mainly through lower hourly wages and different job choices.

### **7.2 What the paper adds to the child-penalty literature**

The paper’s main methodological contribution is to identify the extensive margin of fertility. Many earlier quasi-experimental studies use twins or sex composition, which identify the effect of additional children among women who are already mothers. Lundborg, Plug, and Rasmussen instead identify the effect of becoming a mother. The difference matters because the first child appears to be the career-changing event.

### **7.3 Interpretation of the mechanism**

The results support a dynamic mechanism. In the first two years, mothers reduce hours and participation. After that, they work again, but they do not fully return to the same earnings trajectory. The job-characteristic evidence suggests that women substitute toward jobs closer to home and possibly more compatible with family responsibilities. These jobs pay less, creating a persistent wage penalty.

### **7.4 Policy implications**

The findings imply that paid leave and child care, while important, may not eliminate career costs. Denmark already has generous family policies, yet first childbirth still creates a long-run earnings penalty. Policies that only protect short-run employment may miss the larger long-run margin: wage growth, promotion paths, job flexibility, commuting constraints, and career continuity.

### **7.5 Firm and workplace implications**

The commuting-distance evidence suggests that workplace flexibility has economic value. Firms that can provide flexible work, predictable schedules, remote-work options, or career-compatible part-time arrangements may reduce the need for mothers to switch to lower-paid jobs. The results therefore connect fertility penalties to the design of jobs, not just to household choices.

### **7.6 Equity and human-capital implications**

The large losses among high-earning women imply a substantial loss of high-skill labor-market attachment or wage growth. This matters for gender inequality, leadership pipelines, and the allocation of talent. The child penalty is not just a family issue; it affects the distribution of human capital across firms, occupations, and career ladders.

### **7.7 External validity**

The estimates come from IVF-treated Danish women. These women are older, more educated, and have stronger labor-market attachment than representative mothers. Denmark also has more generous family policy than many other countries. The estimates may therefore understate or differ from effects in countries with weaker family support, but they are internally informative for the treated population.

## **7.8 Research agenda**

Future work should separate the roles of employer policy, household bargaining, job flexibility, and local labor markets. The most important open question is not whether children affect careers, but which institutions prevent temporary childbirth interruptions from becoming permanent wage and job-quality penalties.

## **7.9 Takeaway**

The paper's takeaway is that women can have children and careers, but the first child changes the trajectory of that career in a way that is large, durable, and economically meaningful. The long-run cost is not simply less work; it is lower-paid work.